Can You Defend Your Pass/Fail Decisions?

An Independent Measurement Audit of a Professional Certification Examination

Independent blind analysis of a synthetic professional certification dataset.

Executive conclusion

Correct scoring implementation is established within the supplied records. Substantive validity and operational comparability of certification decisions remain unestablished.

The fictional program can demonstrate that its recorded results follow the supplied key and passing rule. It cannot yet establish that the rule represents the required professional competence, that the forms are operationally interchangeable, or that individual decisions are substantively accurate. Leadership should prioritize the evidence and authority behind the standard, independent item review and external validation before reconsidering operational policy. This is a focused governance agenda, not a declaration of program failure.

What we audited

Forms A and B each contain **60 scored items**, including **20 common items**. Scoring was checked across **2,400 administrations**; the primary item and form comparisons used **2,160 first observed attempts**, with distinct candidates in the form groups. Both forms retain the original **42/60** passing rule. Selected retakes were not treated as independent repeat validation. [1]

The audit examined response patterns, score consistency, item functioning, regional conditional differences, provisional common-item linking and hypothetical decision stability. The analysis used the delivered records and was separated from private simulation-generation information. Convergent methods applied to this dataset do not constitute independent external replication.

Finding 1 / The scoring rule is reproducible

Independent reconstruction found **0 raw-score mismatches and 0 pass/fail mismatches** against the supplied key and rule. This establishes an internally consistent scoring process in these records. It does not independently authenticate the records or validate the key's content, the required competence or the passing standard. Preserve this evidence base while investigating the unresolved questions. [1]

EVIDENCE BOUNDARY

Scoring reproduced

Supplied records follow the supplied key and rule.

Signals to investigate

Item, region and form findings require independent review.

Validity unestablished

Competence criteria and external decision evidence are still needed.

Different signals, different reviews

FINDING 2 / PRIORITY ITEM INVESTIGATIONS

ANC_14

Credit / other-item-total correlation

A -0.3628
B -0.4189

Negative functioning

ANC_14: Credit on this item is associated with lower performance on the rest of the test. Negative functioning appears in both forms across classical item statistics, factor analysis and item-response modeling. Review wording, alternatives, official key rationale, content alignment and delivery conditions. The pattern is strong investigative evidence; it does not prove an incorrect key. [2]

ANC_18

North minus South Percentage points

A 11.51
B 11.11

Regional conditional difference

ANC_18: After allowing for estimated proficiency from other responses, the modeled probability of credit is higher for North than South by **11.51 percentage points in A** and **11.11 in B**, averaged over each form's supported comparison range. These differences require regional interpretation, job-relevance and response-process review. They are not overall pass-rate differences, proven bias or a test of cross-form drift. [3]

B_11

Credit / other-item-total correlation

0.0264

Weak discrimination

B_11: Credit has very weak alignment with performance on the other items, and model-based difficulty is unstable. Review its intended competence, construction and retention rationale. Weak discrimination does not establish that the item is invalid. These different signals warrant distinct expert inquiries rather than automatic deletion or rekeying. [2]

Where the greatest risks remain

The central risk is consequential reliance on an insufficiently evidenced interpretation of passing. Missing standard-setting documentation and authority must be addressed first. This priority concerns evidence verification; it does not establish that the existing standard is wrong.

Item functioning and regional differences require independent substantive review. Form comparability additionally depends on assignment, detailed coverage and stable common-item behavior.
Near-boundary stability depends on response-model and repeat-administration assumptions that have not been externally validated. Potential consequences can be serious while the cause and extent remain uncertain; severity must not be confused with certainty that a defect exists.

Comparing Forms A and B

Finding 3 / Observed outcomes

First-attempt pass rates were **73.40% for A** and **61.94% for B**. These are observed, unadjusted rates from different candidate groups. Form assignment and population comparability have not been independently established, so the gap cannot be attributed solely to form difficulty or treated as proof of unfairness. Obtain assignment, assembly and cohort evidence before drawing a causal conclusion. [4]

Finding 4 / Provisional score correspondence

Under the primary provisional common-item link and the same assumed proficiency population for both forms, an A score of 42 corresponds to approximately **B=38.70**, with a **95% bootstrap interval of 38.15-39.13**. **This is not an authorized conversion or recommended new passing cut.** [5]

RESEARCH CORRESPONDENCE / REFERENCE A=42

B=38.70; 95% bootstrap interval 38.15-39.13

Not an authorized conversion or recommended new passing cut.



B raw-score scale [5]

Conditional on fixed common-item selection, response models and reference population; not an interval for candidate competence.

The result suggests different effective score demands under the adopted models. It remains conditional on the selected common items, response models, population and discrete-score convention. The interval describes parameter and linking uncertainty; it does not cover every content, assignment or model concern. Detailed content and independent anchor validation are still needed. A hypothetical change under this correspondence would not identify a historical certification error.

Decision stability and its limits

Finding 5 / Global and conditional agreement

With the original rule on both forms, modeled **cross-form agreement is 86.59%** across the assumed common proficiency population, with a **95% bootstrap interval of 85.66%-87.64%**. At the fixed modeled proficiency where A's expected score equals 42, cross-form agreement is **46.30%**. This reference is not the group of candidates who obtained an observed score of 42, nor a validated mastery threshold. [6]

TWO CONDITIONING QUESTIONS / ORIGINAL RULES

Population-integrated agreement

86.59%

Across the assumed common proficiency population.



95% bootstrap interval 85.66%-87.64%

Fixed modeled proficiency

46.30%

Where A's expected score equals 42; not the observed raw-42 group or a validated mastery threshold.



Point estimate; parameter uncertainty remains.

Hypothetical independent administrations, unchanged proficiency, no practice or exposure; provisional link and response models. Full agreement scales: 0-100%. [6]

These estimates concern hypothetical independent administrations at unchanged proficiency, without learning, memory or exposure effects. They also assume no unmodeled dependence between item responses; earlier residual screening did not prove that assumption. They are not observed retest agreement. Lower conditional stability matters for candidates close to the consequential boundary; the global average includes people far from it. Bootstrap intervals quantify estimation uncertainty, not the variability of a future individual's outcomes. Real repeated or parallel-form evidence is required before operational reliance.

Finding 6 / Consistency does not establish accuracy

No independently validated competent/not-competent criterion or mastery threshold was supplied. The audit therefore cannot estimate the proportion of substantively correct certifications or identify wrongly accepted or rejected candidates. Model predictions and research-equated scores do not supply ground truth. Obtain the standard-setting and validation evidence before making an accuracy claim. [6]

The next defensible decision

LEADERSHIP SEQUENCE

01 **Verify the basis and authority of the standard.** Obtain the approved competency blueprint, standard-setting rationale, panel evidence, decision authority and validity documentation. Preserve authoritative content, key versions and administration records. Begin this work before new consequential reliance on model-derived rules.

02 **Commission independent content and regional review.** Give experts the original stems, alternatives, key rationales, competency objectives and relevant administration records. Prioritize ANC_14, ANC_18 and B_11, while retaining the targeted monitoring questions in the evidence register. Require a documented explanation of competing interpretations and unresolved disagreements before proposing an item change.

03 **Establish a defensible form-comparison design.** Obtain assembly and assignment procedures, cohort characteristics, delivery conditions and omission causes. Then evaluate common-item coverage and invariance using independent data and an explicitly justified analysis plan. The statistical team needs these inputs before stronger comparability conclusions can be drawn.

04 **Validate decisions prospectively.** Plan independent-cohort and real parallel-form or repeat studies that address practice, exposure, selection and proficiency change. Review predictive performance against the substantive standard. Any future scoring, item or equating policy needs separate authorization after its evidence dependencies are resolved; a favorable model result cannot substitute for that process.

What we cannot yet conclude

The evidence does not establish defective keys, substantive unfairness, an authorized alternative cut, historical certification errors, substantive classification accuracy or operational interchangeability of the forms. Current records and rules remain unchanged. Numerical stability and convergence across methods support qualified investigation, not independent external validation.

The next defensible decision is to approve a bounded evidence-collection and independent-review program, starting with the authority and substantive basis of the passing standard, with explicit review gates before any operational policy is reconsidered.

EVIDENCE NOTES

[1] Scoring-integrity audit: form consistency, observed counts and original-rule checks; synthesis claim on administrative reconstruction.

[2] Classical item statistics, factor loadings and signed item-response diagnostics; synthesis investigations of ANC_14 and B_11.

[3] Within-form primary regional conditional-response results, supported matching ranges and sensitivity qualifications; synthesis investigation of ANC_18.

[4] Observed first-attempt pass-rate tables and form-assignment review; synthesis claim on unadjusted form-group differences.

[5] Primary research score correspondence and paired-bootstrap uncertainty; synthesis claim on equating and integer-boundary uncertainty.

[6] Hypothetical cross-form agreement, cut-region results and uncertainty; synthesis claims on decision stability and unavailable substantive accuracy. The internal evidence map retains exact source cells and issue/action links.